

PROPOSAL: THOUGHTS-AUGMENTED FUNSEARCH FOR ONLINE BIN PACKING HEURISTIC DESIGN

Li Mengxiao, Ye Hengrui & Jiang Yuchen

Department of Computer Science, City University of Hong Kong

{mengxli2-c, hengruiye2-c, yjiang456-c}@my.cityu.edu.hk

1 GROUP MEMBERS

Our group consists of: **Li Mengxiao**, **Ye Hengrui**, and **Jiang Yuchen**.

2 PROBLEM SELECTION AND RATIONALE

We choose **Topic 2a: Thoughts-augmented FunSearch** for the **Online Bin Packing Problem**. In this task, items arrive sequentially and must be immediately assigned to bins of fixed capacity to minimize total usage.

Rationale: Traditional FunSearch evolves raw code, which often lacks high-level strategic reasoning. We propose adding natural language *“thoughts”* before code generation to activate the LLM’s **Chain-of-Thought (CoT)** capabilities. This allows the model to conceptualize complex strategies (e.g., adaptive capacity management) before implementation, leading to more robust heuristics and faster convergence than raw code mutation.

3 TENTATIVE PLAN

We will implement an enhanced evolutionary loop based on the project template:

- **Thought-Augmented Mutation:** The LLM will be prompted to first generate a reasoning *“Thought”* (strategic intent) and then the corresponding Python code.
- **Program Database:** We will store (Thought, Code) pairs, allowing the LLM to retrieve and refine successful logical patterns rather than just raw code snippets.
- **Feedback Loop:** Heuristics will be evaluated in the provided sandbox. The scores will reinforce both the successful *“thoughts”* and their code implementations.

4 EVALUATION PLAN

We will evaluate whether *“thoughts”* improve heuristic discovery via the following setup:

- **Datasets:** We will use **OR-Library** and **WeiBull** datasets, covering diverse item distributions to ensure the generalizability of our heuristics.
- **Baselines:** We will compare our results against (1) **Vanilla FunSearch** (code-only) and (2) **Human Heuristics** (Best-Fit and Least Remaining Capacity).
- **Metrics:** The primary metric is the **number of bins used**. We also measure **search efficiency** by tracking the iterations required to outperform baselines.